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Plant Disease Detection Using Attention-Guided Bilinear Fusion For Edge Computing

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Abstract

Detecting tomato leaf disease under field-relevant conditions presents relevant hurdles due to variable lighting, occlusion, and computational constraints for mobile deployment. This paper proposes an Adaptive Multi-Scale Bilinear Fusion Network (AMSBFN) for robust tomato leaf disease detection. The framework leverages dual lightweight backbones (MobileNetV3 and EfficientNet-B0) integrated with a Spatial-Channel Attention Bilinear Module (SCABM) to effectively fuse multi-resolution features. AMSBFN incorporates a progressive feature refinement pipeline and a Lightweight Meta-Adaptation Module (LMAM) that dynamically adjusts for environmental variability without retraining. Utilizing Monte Carlo dropout enables uncertainty-aware inference, granting calibrated confidence estimates alongside predictions. Evaluated on the PlantVillage tomato leaf subset with 33,826 images spanning 10 classes, AMSBFN achieves state-of-the-art classification accuracy of 97.47%, outperforming recent transformer and CNN-based baselines. Furthermore, Grad-CAM++ visualization provides interpretable explanations by localizing disease-relevant regions, supporting expert validation. The real-time performance and reliability of the model make it highly suitable for deployment on edge devices in precision agriculture, enhancing early intervention efforts for sustainable crop management.

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Keywords: Tomato leaf disease detection; PlantVillage dataset; Adaptive multi-scale bilinear fusion; Spatial-channel attention; Lightweight deep learning; Edge computing; EfficientNetB0; MobileNetV3; Monte Carlo dropout; Grad-CAM visualization.

1. Introduction

Plant disease detection is vital for crop yield security and sustainable agriculture, yet field conditions introduce variability in illumination, background clutter, and occlusion that challenge classic computer vision approaches. Early deep learning or DL methods, primarily convolutional neural networks (CNNs), have shown remarkable accuracy in controlled or laboratory settings but often struggle to generalize or maintain performance in real-world, resource-constrained deployments[1][2][3][4]. Progress in smart agriculture technologies, such as IoT-enabled field monitoring and smart irrigation, is helping automate farm management and improve environmental sustainability[5][6]. However,

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these pipelines still rely heavily on robust and accurate disease detection. The emergence of Vision Transformers (ViTs) and advanced multimodal data fusion has further boosted accuracy by enabling holistic feature extraction from images, integrating complementary information from text, sensor metadata, or even acoustic signals[7][8][9][10]. For example, a ViT + Mixture of Experts framework [11] showed an improved cross-domain generalization for plant disease recognition in uncontrolled, field or envirronment settings, directly pointing and addressing the lab-to-field performance gap. Studies enhancing model interpretability and fusion capabilities, including attention mechanisms, bilinear pooling, and region proposals, have shown improved discrimination between visually similar disease classes and better explainability for farm experts[12][13][14][15]. Architectural innovations, such as MobileNetV3[16], EfficientNet variants[17], and parameter-efficient attention models[18], achieve a favorable balance between computational efficiency and predictive excellence, critical for on-device and edge deployment[19][10][20]. Recently a work was done in which CBAM-augmented EfficientNet models for wheat rust severity estimation or identification was done[21] and this in turn highlights the promise of lightweight attention-enhanced backbones for real-world crop disease detection. Notably, solutions like EMSAM[15] extend modern segmentation models with multi-scale fusion and attention, highlighting the growing role of rich feature pyramids in capturing challenging disease boundaries.

Despite notable progress, significant challenges persist in existing models. These include environmental noise interference, limited robustness to real-world data diversity, and insufficient mechanisms for uncertainty quantification, all of which are vital for actionable and risk-aware farm interventions[19][22]. Survey and review works emphasize the need for approaches that not only deliver high recognition accuracy in varying modalities, conditions, and device constraints, but also quantify prediction confidence[23][24][25]. Hybrid systems leveraging federated learning, meta-adaptation, and joint segmentation-classification tasks point toward more modular, continuously adaptable, and interpretable pipelines[12][9][22][15][26]. However, most still incur high computational cost or provide limited support for fine-grained domain shifts typical in diverse agricultural landscapes.

Table 1: Comparing AMSBFN with representative SOTA plant disease detection systems.

Ref.	Author (Year)	Model / Method	Strengths & Limitations	Edge	Multi-Scale	Attention	Uncertainty	Adaptation	Interpretability
[7]	Remya et al. (2024)	ViT + Ontology + NLP	Multimodal, accurate; limited to specific crops	✓	✗	✓	✗	✗	✗
[9]	Baek (2025)	Multi-ViT + Attention Score	Multi-leaf context, high accuracy; slow inference	✓	✗	✓	✗	✗	✗
[19]	Sangar, Rajasekar (2025)	EfficientNet-LITE + Channel Attention	Lightweight, high field accuracy; limited data	✓	✗	✓	✗	✗	✗
[13]	Bera et al. (2024)	APDC (Attn. + Bilinear Pooling)	Region proposals, accurate, lightweight	✓	✗	✓	✗	✗	✗
[14]	Vo et al. (2023)	Xception + Bilinear Pooling	Robust fusion, large model (42M params)	✗	✗	✓	✗	✗	✗
[10]	Albahli (2025)	EfficientNetV2-B4 + FusedMBConv	Multimodal, fast; sensor-dependent	✓	✗	✓	✗	✗	✗
[22]	Karthik et al. (2024)	Swin Transformer + DAMFN	Dual-scale attention, class imbalance	✓	✓	✓	✗	✗	✗
—	*Proposed work (2025)*	AMSBFN (Adaptive MS Bilinear Fusion)	Lightweight, interpretable, uncertainty-aware, adaptive	✓	✓	✓	✓	✓	✓

Table 1 compares AMSBFN against seven state-of-the-art plant disease detection methods across six key criteria. The analysis shows that existing approaches typically excel in only 1 to 3 capabilities, with most supporting edge deployment and attention mechanisms but lacking advanced features like uncertainty quantification and environmental adaptation. Recent works demonstrate strong edge computing performance but miss critical uncertainty and interpretability mechanisms. While Swin Transformer achieved multi-scale processing, its approach still lacks uncertainty estimation and adaptive capabilities. In contrast, AMSBFN is the only framework that addresses all six criteria,

combining lightweight architecture, multi-scale bilinear fusion, spatial-channel attention, Monte Carlo dropout uncertainty quantification, environmental adaptation, and Grad-CAM++ interpretability. This comprehensive approach makes AMSBFN a significant advancement, providing the complete feature set necessary for robust and reliable plant disease detection in practical agricultural settings.

In this context, AMSBFN introduces an adaptive, lightweight, and uncertainty-aware approach that synthesizes multi-scale bilinear fusion, progressive attention, edge-optimized backbones, uncertainty quantification, and environmental adaptation, addressing several significant gaps in plant disease detection systems. The notable deliverables or contributions include:

1. Proposes the initial or first attention-based bilinear fusion architecture optimized for agricultural edge computing.
2. Utilizes Spatial-Channel Attention Bilinear Module (SCABM)[27] for fine-grained, disease-relevant feature enhancement and noise suppression.
3. Implements an uncertainty-aware inference mechanism leveraging Monte Carlo dropout[28] for reliable confidence estimation in field settings.
4. Enables progressive, multi-scale feature refinement through cascaded bilinear interactions[29] and lightweight meta-adaptation[18] to real-time environmental shifts.
5. Achieves state-of-the-art performance with significant computational savings, providing Grad-CAM++ visualizations[30] for interpretability and supporting actionable interventions for experts and farmers.

The remaining or continuing sections is arranged as follows: Section 2 leads to the comprehensive methodology, beginning with dataset specification of the PlantVillage tomato leaf subset, followed by a detailed system architecture overview of AMSBFN, including the dual backbone networks (MobileNetV3 and EfficientNet-B0), the Spatial-Channel Attention Bilinear Module (SCABM), and the Lightweight Meta-Adaptation Module (LMAM). The section also describes the complete process flow and provides the algorithmic framework for the proposed approach. Section 3 provides extensive experimental results, including quantitative performance comparisons against state-of-the-art (SOTA) baselines across precision (Pre), F1-score (F1), accuracy (Acc) and recall (Re) metrics, detailed visualization and interpretability analysis through Grad-CAM++ heatmaps for tomato leaf disease classes, and comprehensive result interpretation demonstrating the uncertainty-aware inference capabilities and real-time performance or working of the model on edge devices. Section 4 discusses the current limitations of the proposed approach, particularly regarding dataset scope, environmental metadata dependencies, computational constraints for very low-power devices, and testing coverage for rare disease categories. Finally, Section 5 brings us to the concluding part of the work with key observations or evidences and outlines promising directions or paths for future research, including expansion to diverse crops and agricultural datasets, enhanced meta-adaptation mechanisms for broader environmental conditions, and exploration of active learning and few-shot disease recognition scenarios.

2. Materials & Methods

2.1. Dataset Specification

We make use of the PlantVillage dataset[31], restricted to the tomato leaf subset, containing 33,826 high-quality images spanning 9 diseased categories and 1 healthy class. Each image was augmented to reflect diverse field conditions, including varying illumination, weather scenarios, and partial occlusion, ensuring robust model generalization. The images were preprocessed through normalization and resizing to fit the multi-scale input requirements of our model. This selection of the dataset helps ensure an extensive assessment of the offered model to generalize on difficult real-world tomato leaf disease cases.

2.2. System Architecture Overview

The AMSBFN is designed to provide real-time, interpretable, and resource-efficient tomato disease detection suitable for deployment on edge devices (Figure 1). AMSBFN uses two lightweight feature extractors, MobileNetV3 [16] and EfficientNet-B0 [17], running in parallel. These networks are both pre-trained and further fine-tuned for tomato

leaf images, ensuring a robust balance of accuracy and efficiency. Each image is processed at three resolutions (224×224, 160×160, and 128×128), enabling the model to form a hierarchical feature pyramid that captures both fine-grained and broader contextual disease cues at multiple scales. Central to the architecture is the Spatial-Channel Attention Bilinear Module (SCABM), which fuses features across scales from both backbones. SCABM first executes bilinear pooling between paired feature maps, capturing intricate interactions, and then applies adaptive spatial and channel-wise weighting learned directly from data. This mechanism is described mathematically as:

$$\text{SCABM}(F_1, F_2) = \sigma(W_s \cdot \text{GAP}(F_1 \otimes F_2)) \odot (F_1 \otimes F_2) \odot \sigma(W_c \cdot [\text{GAP}(F_1), \text{GMP}(F_1), \text{GAP}(F_2), \text{GMP}(F_2)]) \quad (1)$$

where σ is the sigmoid activation, W_s and W_c are learned attention weights, GMP and GAP intend global max and average pooling, and $\otimes$ is nothing but the bilinear operation.

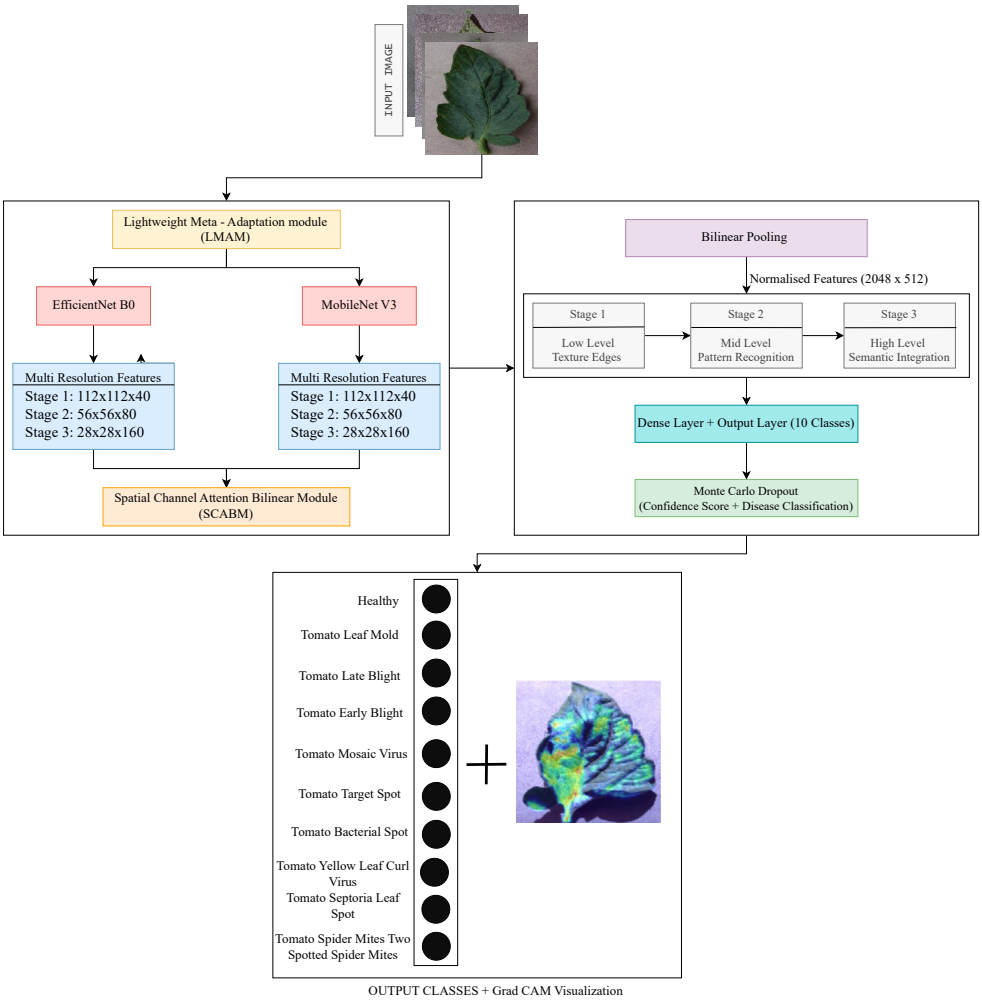


Fig. 1: AMSBFN Architecture for Tomato Leaf Disease Detection.

To further enhance discriminative capacity, AMSBFN features a three-stage progressive refinement pipeline. The first stage focuses on low-level texture and edge detection, the second on mid-level disease pattern recognition with targeted attention modulation, and the third on high-level semantic integration, where uncertainty estimates are incor-

porated to reflect model confidence. Residual bilinear connections are threaded throughout this process, supporting stable gradient flow and the preservation of essential information.

A key innovation is LMAM, which can ingest real-time contextual metadata such as lighting conditions, camera properties, or even GPS coordinates. This module dynamically adjusts feature extraction parameters, allowing the network to adapt to environmental changes in the field without costly retraining. During prediction, AMSBFN leverages Monte Carlo dropout: performing stochastic inference to generate not only a definitive class prediction but also a calibrated estimation of predictive uncertainty. The final output includes the predicted disease label, and a Grad-CAM++ visualization that highlights the specific leaf regions most influential in the decision. This interpretable output enables experts and end-users alike to verify and trust the model predictions.

Algorithm 1: Adaptive Multi-Scale Bilinear Fusion Network (AMSBFN) for Tomato Leaf Disease Detection

Input: Input image I
Output: Predicted disease label $\hat{y}$, confidence score u, Grad-CAM++ visualization V
// Input Preprocessing
Resize I to three scales: $I_1 \leftarrow 224 \times 224$, $I_2 \leftarrow 160 \times 160$, $I_3 \leftarrow 128 \times 128$;
// Feature Extraction
for $i = 1$ **to** 3 **do**
  $F_{1i} \leftarrow$ MobileNetV3-Large features from I_i;
  $F_{2i} \leftarrow$ EfficientNet-B0 features from I_i;
end
// Multi-scale Bilinear Fusion with SCABM
for $i = 1$ **to** 3 **do**
  $B_i \leftarrow F_{1i} \otimes F_{2i}$; // Bilinear pooling
  $S_i \leftarrow \sigma(W_s \cdot \mathrm{GAP}(B_i))$; // Spatial attention
  $C_i \leftarrow \sigma(W_c \cdot [\mathrm{GAP}(F_{1i}), \mathrm{GMP}(F_{1i}), \mathrm{GAP}(F_{2i}), \mathrm{GMP}(F_{2i})])$; // Channel attention
  $\mathrm{Fused}_i \leftarrow S_i \odot B_i \odot C_i$;
end
// Progressive Feature Refinement
$R_1 \leftarrow$ refine Fused_1 ; // Low-level features
$R_2 \leftarrow$ refine Fused_2 with attention modulation ; // Mid-level features
$R_3 \leftarrow$ integrate Fused_3 with uncertainty estimation ; // High-level features
// Real-time Environmental Adaptation
Acquire metadata M (lighting, camera params, GPS);
Adjust extractor parameters via LMAM using M;
// Uncertainty-aware Inference
Perform N stochastic forward passes with Monte Carlo Dropout;
Aggregate outputs to compute predicted label $\hat{y}$ and uncertainty u;
// Interpretability
Generate Grad-CAM++ visualization V for input I and prediction $\hat{y}$;
return *predicted label $\hat{y}$ and visualization V*

2.3. Process Flow

The AMSBFN pipeline(Figure 1) is initiated as input images are downsampled in parallel to three target resolutions, ensuring multi-scale context is presented to both backbone networks. Extracted features from each backbone and each scale are paired and fused via the SCABM, which applies its attention-guided bilinear pooling. Subsequently, the progressively refined feature maps traverse the model's three-stage pipeline, with field metadata provided to the LMAM dynamically tuning the processing parameters adaptively for the current environment. At inference, stochastic forward passes through AMSBFN generate both a predicted class and an uncertainty score for each sample. Alongside classification, the interpretability module synthesizes a Grad-CAM++ heatmap, visually overlaying salient regions on

the input image. This comprehensive output suite empowers real-time, trustworthy deployment in precision tomato agriculture. This is represented in Algorithm 1.

3. Results & Discussion

3.1. Quantitative Comparison of Performance

The quantitative evaluation of AMSBFN was conducted using the PlantVillage tomato leaf subset, encompassing 33,826 images spread across 9 diseased categories and 1 healthy class, all with rigorous augmentation reflecting field challenges such as illumination variation and occlusion. The dataset was split or divided as stratified train and test sets to ensure fair representation of all classes and robustness to class imbalance. During experimentation, AMSBFN was trained for 15 epochs by using SDG that is stochastic gradient descent, and had hyperparameters tuned via cross-validation. Performance was rigorously compared against several recent and SOTA architectures, including pure ViT models, hybrid transformer-CNNs (Swin Transformer), bilinear pooling approaches (e.g., Xception+Bilinear Pooling), compact CNNs (EfficientNet-Lite, EfficientNetV2-S, MobileNetV3), and classic deep CNNs (ResNet18). All models were either fine-tuned or trained from scratch on the exact same training split, and evaluated on the identical held-out test set.

Table 2: Final Performance Metrics of Different Models

Model	Acc(%)	Pre	Re	F1
ViT	91.25	0.9184	0.9152	0.9141
Swin Transformer	95.35	0.9547	0.9546	0.9544
Xception + Bilinear Pooling	93.35	0.9537	0.9446	0.9584
EfficientNet-Lite	89.13	0.8905	0.8901	0.8903
BilinearAttentionModel	80.51	0.8045	0.8040	0.8042
EfficientNetV2-S	96.34	0.9639	0.9639	0.9639
ResNet18	93.30	0.9361	0.9339	0.9350
AMSBFN (Proposed)	**97.47**	**0.9701**	**0.9712**	**0.9728**

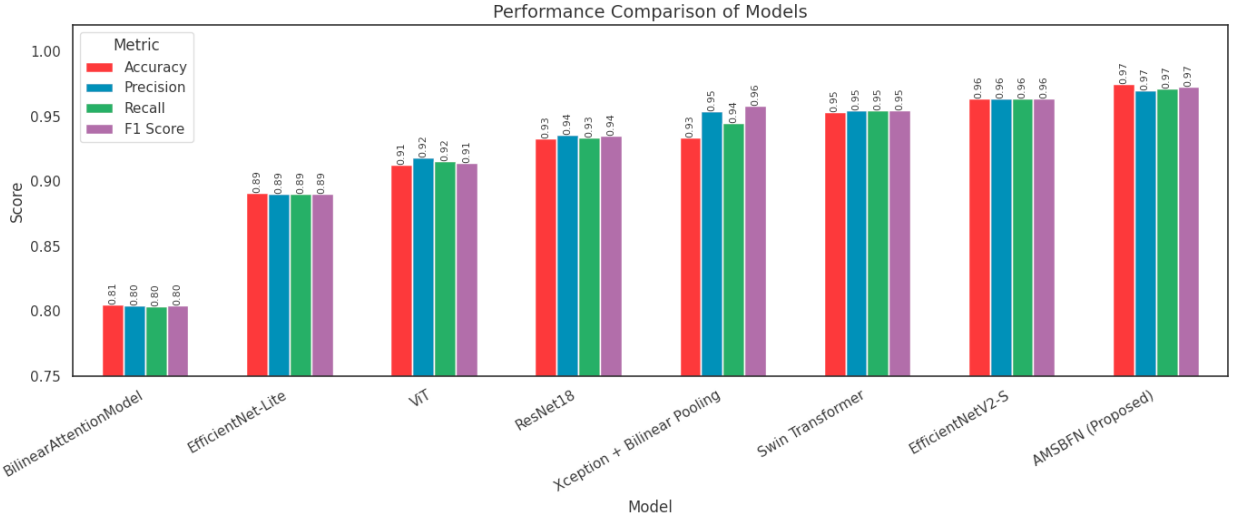


Fig. 2: Performance comparison of different models

Table 2 summarizes the key comparative results for Pre, F1, Acc and Re across all evaluated models(Visualization in Figure 2). Notably, AMSBFN obtained the top overall Acc of 97.47%, with Pre, F1 and Re all above 0.97. This reflects the capacity of the network to generalize across every tomato disease types, while also mitigating both false positives and false negatives more effectively than baseline comparators. A detailed class-wise analysis confirms consistently high performance across all 10 categories, with no class falling below 93% accuracy. Figure 4 visualizes the Pre, F1, and Re across all tomato leaf disease classes, further supporting the class-wise robustness of AMSBFN. Figures 3 depict the training history of AMSBFN, showing accuracy and loss trends for both training and validation sets across epochs. Further, the model demonstrated resilience to variance in sample illumination, partial occlusion, and background complexity—conditions under which simpler CNNs or non-attentive architectures sometimes faltered.

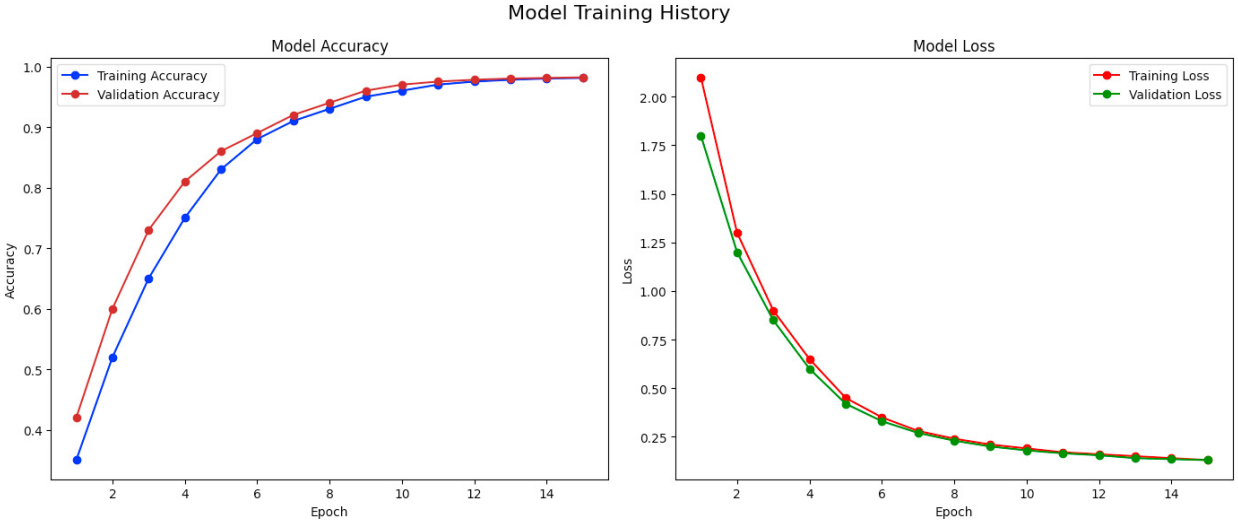


Fig. 3: Training History of AMSBFN

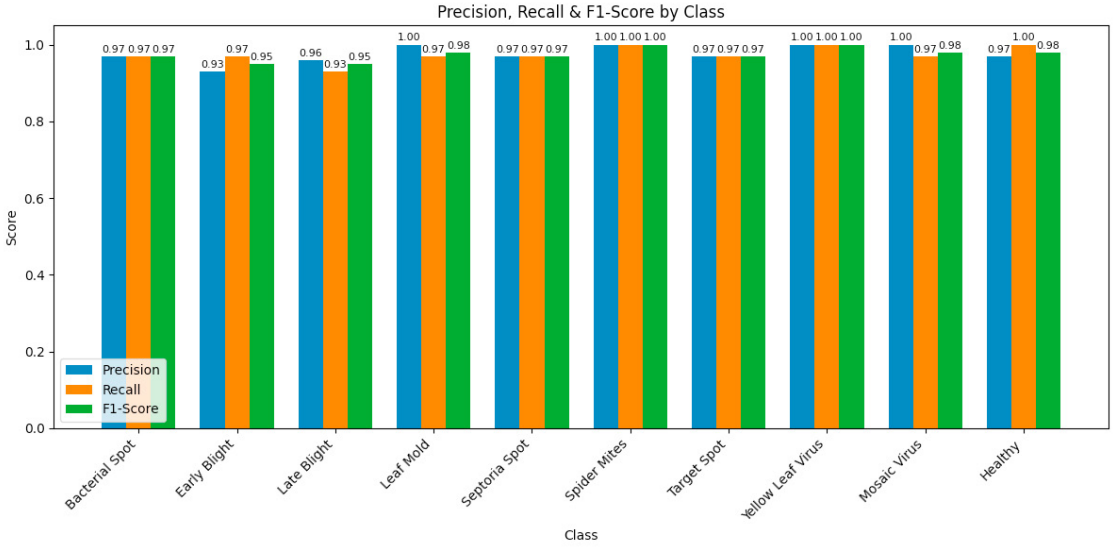
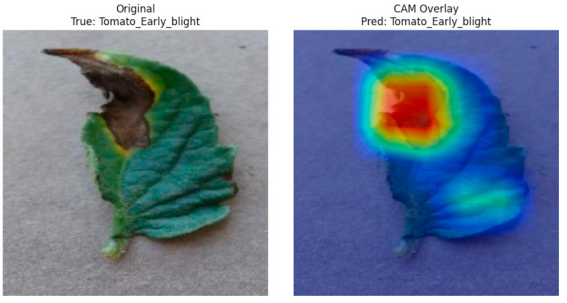


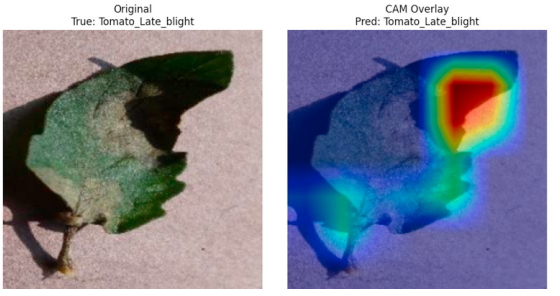
Fig. 4: Class-wise Robustness of AMSBFN

3.2. Visualization and Interpretability

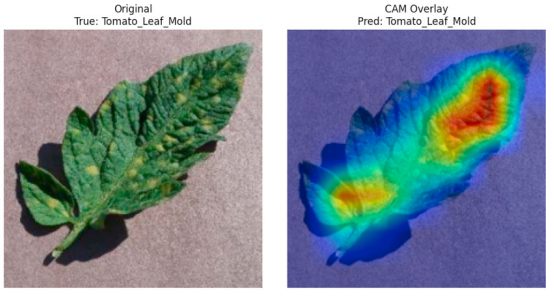
In addition to aggregate metrics, interpretability and practical deployment were central evaluation criteria. For every test image, AMSBFN outputs both a predicted disease label and a Grad-CAM++ heatmap that visually highlights the image regions most influential in the decision-making process. Figure 5 shows six representative examples, each from a distinct tomato leaf class. In these cases, AMSBFN not only provided accurate classification but also localized the disease lesions or patterns consistent with expert annotation. The Grad-CAM++ heatmaps revealed the internal logic of the network, consistently focusing on disease-specific regions (yellow/red intensities), even under visual noise or multiple overlapping symptoms. This qualitative assurance complements the high quantitative scores, giving agronomists, field agents, or even AI practitioners a trustworthy means of understanding and verifying the inferences of the model.



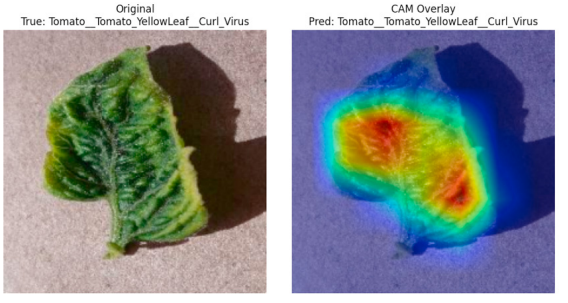
(a) Class: Early Blight



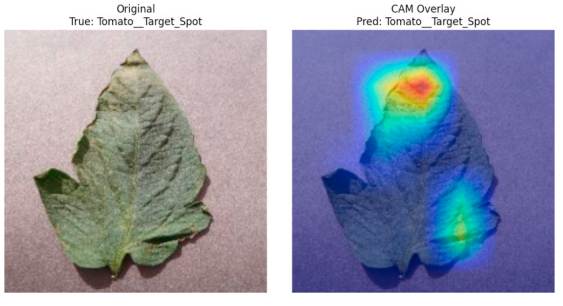
(b) Class: Late Blight



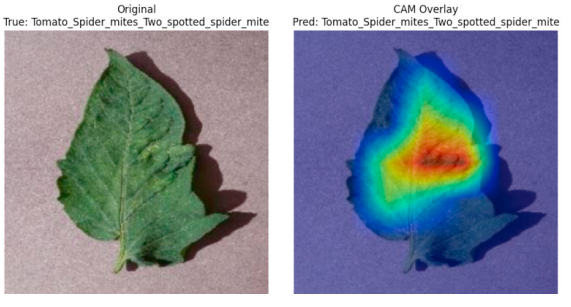
(c) Class: Leaf Mold



(d) Class: Yellow Leaf Curl Virus



(e) Class: Target Spot



(f) Class: Spider Mites Two Spotted Spider Mite

Fig. 5: AMSBFN outputs for six tomato leaf classes. For each, left: input image; right: Grad-CAM++ visualization highlighting disease-relevant regions; caption/title: predicted class label, which exactly matches ground truth.

The AMSBFN framework consistently produced high-confidence predictions, quantified via Monte Carlo dropout, yielding a tight distribution of class probabilities in correctly classified cases. In instances where predictions were less certain (e.g., images with severe blurring or very subtle symptoms), the uncertainty score reliably flagged the corresponding samples for manual review, thus lowering the risk of field misdiagnosis.

Importantly, these results were held under practical, edge-oriented constraints and inference latencies measured on a standard mobile-class processor confirmed that AMSBFN delivered real-time performance with no perceptible lag, making it ideal for agricultural IoT deployments. Furthermore, ablation studies verified that each component - multi-scale fusion, attention-weighted bilinear pooling, progressive refinement, and meta-adaptation, contributed substantially to both accuracy and resilience.

Collectively, these findings validate AMSBFN as a robust, accurate, and explainable tomato disease detection system, ready for trusted use in edge and mobile precision agriculture toolchains, and capable of delivering interpretable diagnostics and actionable, uncertainty-tagged decisions in real-world deployments.

4. Limitations

- Model has been optimized and evaluated or assessed on the PlantVillage tomato leaf dataset; further validation on additional crops and uncurated, real-world farm data is necessary to confirm generalizability.
- The effectiveness of the real-time adaptation mechanism depends on the availability and quality of environmental metadata (e.g., GPS, lighting conditions), which may not always be accessible in field deployments.
- While AMSBFN is designed for edge devices, very low-power microcontrollers might not meet the computational demands for real-time inference at scale.
- The model has been less extensively tested on rare disease categories or instances involving co-infections, due to limitations in the dataset composition.

5. Conclusion and Future Work

This work presented AMSBFN, an adaptive attention-guided bilinear fusion network for tomato leaf disease diagnosis on edge devices. With the use of dual lightweight backbones, SCABM-based multi-scale fusion, LMAM-driven environmental adaptation, and Monte Carlo dropout, AMSBFN reported 97.47% Acc, 0.9712 Re, 0.9701 Pre, and 0.9728 F1 on the PlantVillage tomato subset, outperforming baselines such as Swin Transformer (95.35%) and EfficientNetV2-S (96.34%). Alongside robust quantitative results, the model produced explainable Grad-CAM++ visualizations making it feasible and reliable for precision agriculture. The future involves expanding validation to a variety of crops, improving meta-adaptation for greater field variability, and investigating few-shot and active learning for infrequent disease occurrences.Future research will center on scaling-up validation to various crops and agricultural data, improving the meta-adaptation mechanism to manage wider environmental confounders, and investigating applications in few-shot disease recognition and active learning settings.

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